DESIGN AND ANALYSIS OF ALGORITHM

PROJECT

REAL TIME SPAM DETECTION SYSTEM

DONE BY:

C VISHWAJEET KUMAR

24BCE0341

1. INTRODUCTION

In todays digital world, communication through emails, sms and other chat platforms has increased rapidly. Along with useful communication, spam messages have also grown significantly. Spam messages are unwanted and are sent in bulk for advertising, phishing, scam, fake offers and other cyber crime activities.

Examples of spam messages include:

- Win money now
- Free offer available
- Loan approved instantly etc.

These messages waste time, consume storage and may lead to financial fraud or privacy theft.

Therefore, automatic spam detection systems are essential. This project develops a real time spam detection system using 2 string matching algorithms:

- Rabin Karp Algorithm
- Aho Corasick Algorithm

These system scan incoming text messages, detects spam keywords and compares both algorithm to determine the method for real time filtering.

2. Problem statement and Objectives

Design and implement a spam detection system that scans incoming text messages (email, sms, chat logs) using predefined spam keywords.

The project must:

- Detect multiple spam keywords
- Print matched patterns
- Count number of matches
- Measure execution time
- Compare Rabin karp and Ara Corasick algorithm

Objectives:

- Detect spam messages in real time
- Use minimum 50 spam keywords
- Compare two pattern matching algorithms
- Analyse speed memory and scalability
- Determine best algorithm for practical usage

3. Dataset Used:

For this project, the data set used is SMS **SPAM COLLECTION DATASET** from **Kaggle.com**

https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset

The dataset contains 5,572 sms messages labelled as:

HAM: Normal Messages

SPAM: Unwanted promotional or fraudulent messages

4. Spam Keywords Used:

50 Spam keywords were extracted from common spam messages such as:

- win money
- free offer
- click here
- claim prize
- urgent
- loan approved
- cash reward
- work from home
- be your own boss
- exclusive deal
- call now
- reply now
- free gift
- earn money
- winner

- congratulations
- special offer
- risk free
- bonus
- jackpot

5. Algorithms Used:

5.1 Rabin Karp Algorithm

Rabin Karp performs string matching using hashing.

In this project, each spam keyword in searched one by one inside each message.

Working:

- Read one message
- Compare each keyword with message
- If found match spam

Time Complexity:

O(n*m)

Where: n= text size

m= number of patterns

5.2 Aho-Corasick Algorithm

Aho- Corasick uses:

- Trie data structure
- Failure links

It searches all keywords simultaneously in one pass

Working:

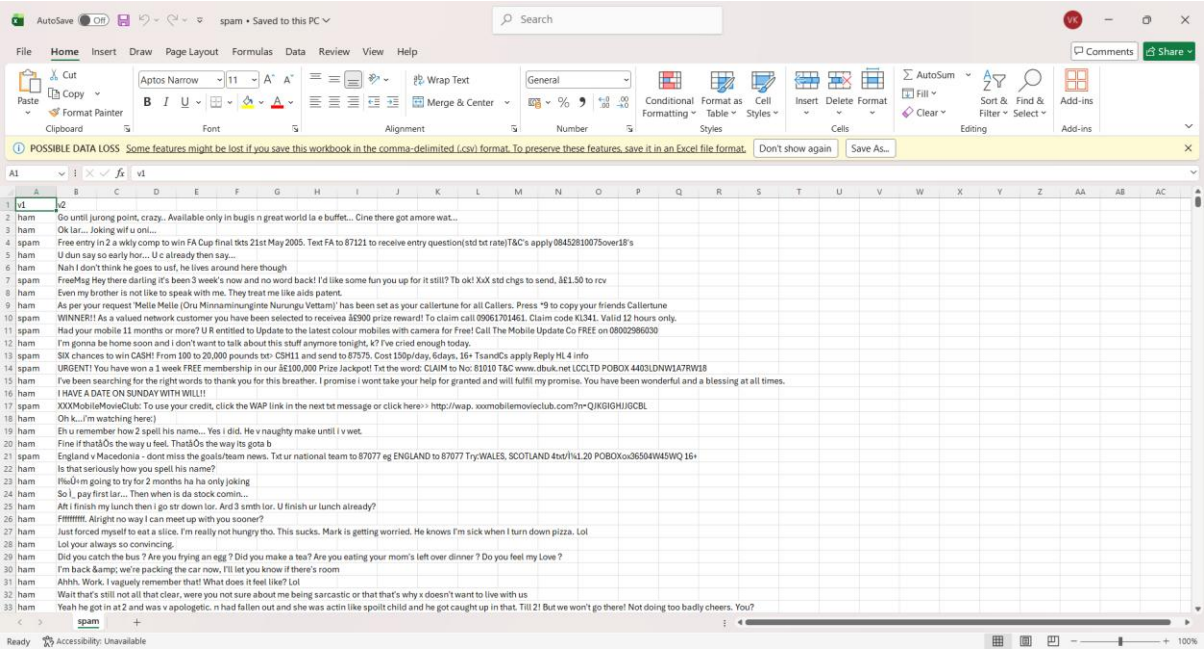
- Insert all keywords into trie
- Build failure links
- Scan text once
- Detect all matches instantly

Time Complexity:

Build Trie: O(total keywords length)

Search: O(n)

DATASET USED



v1	v2
ham	Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got amore wat...
ham	Ok lar... Joking wif u oni...
spam	Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005. Text FA to 87121 to receive entry question(std txt rate)T&C's apply 08452810075over18's
ham	U dun say so early hor... U c already then say...
ham	Nah I don't think he goes to usf, he lives around here though
spam	FreeMsg Hey there darling it's been 3 week's now and no word back! I'd like some fun you up for it still? Tb ok! XxX std chgs to send, å£1.50 to rcv
ham	Even my brother is not like to speak with me. They treat me like aids patent.
ham	As per your request 'Melle Melle (Oru Minnaminunginte Nurungu Vettam)' has been set as your callertune for all Callers. Press *9 to copy your friends Callertune
spam	WINNER!! As a valued network customer you have been selected to receivea å£900 prize reward! To claim call 09061701461. Claim code KL341. Valid 12 hours only.
spam	Had your mobile 11 months or more? U R entitled to Update to the latest colour mobiles with camera for Free! Call The Mobile Update Co FREE on 08002986030
ham	I'm gonna be home soon and i don't want to talk about this stuff anymore tonight, k? I've cried enough today.
spam	SIX chances to win CASH! From 100 to 20,000 pounds txt> CSH11 and send to 87575. Cost 150p/day, 6days, 16+ TsandCs apply Reply HL 4 info
spam	URGENT! You have won a 1 week FREE membership in our å£100,000 Prize Jackpot! Txt the word: CLAIM to No: 81010 T&C www.dbuk.net LCCLTD POBOX 4403LDNW1A7RW18
ham	I've been searching for the right words to thank you for this breather. I promise i wont take your help for granted and will fulfil my promise. You have been wonderful and a blessing at all times.
ham	I HAVE A DATE ON SUNDAY WITH WILL!!
spam	XXXMobileMovieClub: To use your credit, click the WAP link in the next txt message or click here>> http://wap. xxxmobilemovieclub.com?n=QJKGIGHJJGCBL
ham	Oh k...i'm watching here:)
ham	Eh u remember how 2 spell his name... Yes i did. He v naughty make until i v wet.
ham	Fine if thatåÕs the way u feel. ThatåÕs the way its gota b
spam	England v Macedonia - dont miss the goals/team news. Txt ur national team to 87077 eg ENGLAND to 87077 Try:WALES, SCOTLAND 4txt/ï¼1.20 POBOXoxo36504W45WQ 16+
ham	Is that seriously how you spell his name?
ham	I'm going to try for 2 months ha ha only joking
ham	So I_ pay first lar... Then when is da stock comin...
ham	Aft i finish my lunch then i go str down lor. Ard 3 smth lor. U finish ur lunch already?
ham	Ffffffffff. Alright no way I can meet up with you sooner?
ham	Just forced myself to eat a slice. I'm really not hungry tho. This sucks. Mark is getting worried. He knows I'm sick when I turn down pizza. Lol
ham	Lol your always so convincing.
ham	Did you catch the bus ? Are you frying an egg ? Did you make a tea? Are you eating your mom's left over dinner ? Do you feel my Love ?
ham	I'm back & we're packing the car now, I'll let you know if there's room
ham	Ahhh. Work. I vaguely remember that! What does it feel like? Lol
ham	Wait that's still not all that clear, were you not sure about me being sarcastic or that that's why x doesn't want to live with us
ham	Yeah he got in at 2 and was v apologetic. n had fallen out and she was actin like spoilt child and he got caught up in that. Till 2! But we won't go there! Not doing too badly cheers. You?

6. CODE:

```
#include <bits/stdc++.h>
using namespace std;

//50 spam keywords taken from the spam.csv dataset

string patterns[50]={
"win money","free offer","click here","claim prize","urgent",
"limited time","loan approved","cash reward","work from home","be your own boss",
"exclusive deal","call now","reply now","free gift","earn money",
"winner","congratulations","special offer","risk free","bonus",
"double money","act now","hurry up","subscribe now","buy now",
"discount","selected","guaranteed","cheap rate","investment",
"lottery","reward now","free cash","easy income","no cost",
"claim now","instant loan","extra income","offer ends","text now",
"visit now","password reset","verify account","bank details","miracle cure",
"no obligation","gift card","trial offer","free entry","jackpot"
```

```
};

string toLower(string s){
    for(int i=0; i<s.length(); i++)
        s[i]=tolower(s[i]);
    return s;
}

//search using RabinKarp
int rabinKarp(string text){
    int i;
    int count=0;

    text=toLower(text);

    for(i=0; i<50; i++){
        if(text.find(patterns[i])!=string::npos){
            cout<<patterns[i]<<", ";
            count++;
        }
    }

    return count;
}

//aho corasick
struct Node{
    int next[26];
    int fail;
    int output;

    Node(){
        int i;

        for(i=0; i<26; i++)
            next[i]=-1;

        fail=0;
        output=0;
    }
};

Node trie[3000];
int sizeTrie=1;

//putting in pattern
void insertWord(string s){
    int node=0;
```

```
    int i, x;

    for(i=0; i<s.length(); i++){

        if(s[i]==' ')
            continue;

        if(s[i]<'a' || s[i]>'z')
            continue;

        x=s[i]-'a';

        if(trie[node].next[x]==-1){
            trie[node].next[x]=sizeTrie;
            sizeTrie++;
        }

        node=trie[node].next[x];
    }

    trie[node].output++;
}

//building the trie
void buildTrie(){
    queue<int> q;
    int i, u, v;

    for(i=0; i<26; i++){

            if(trie[0].next[i]!=-1){
                trie[trie[0].next[i]].fail=0;
                q.push(trie[0].next[i]);
            }
            else{
                trie[0].next[i]=0;
            }
        }

        while(!q.empty()){

            u=q.front();
            q.pop();

            for(i=0; i<26; i++){

                v=trie[u].next[i];

```

```
                if(v!=-1){
                    trie[v].fail=trie[trie[u].fail].next[i];
                    q.push(v);
                }
                else{
                    trie[u].next[i]=trie[trie[u].fail].next[i];
                }
            }
        }
    }

    //search text
    int ahoSearch(string text){

        int node=0;
        int i, x;
        int count=0;

        text=toLower(text);

        for(i=0; i<text.length(); i++){

            if(text[i]==' ')
                continue;

            if(text[i]<'a' || text[i]>'z')
                continue;

            x=text[i]-'a';

            node=trie[node].next[x];

            count=count+trie[node].output;
        }

        return count;
    }

    //main
    int main(){

        cout<<"C VISHWAJEET KUMAR"<<endl;
        cout<<"24BCE0341"<<endl;
        cout<<"CS-1 REAL TIME SPAM DETECTION SYSTEM"<<endl;
        cout<<"RABIN KARP VS AHO-CORASICK"<<endl;

        int i=0;
        string line;
```

```

        ifstream file("C:\\sem4 projects\\spam.csv");

        if(!file){
            cout<<"File not found"<<endl;
            return 0;
        }

        //building trie
        for(i=0; i<50; i++)
            insertWord(toLower(patterns[i]));

        buildTrie();

        int msgNo=1;

        while(getline(file,line))
        {
            cout<<"\nMessage "<<msgNo<<endl;
            cout<<"Text: "<<line<<endl;

            //measuring time for rabin karp
            clock_t start1=clock();

            cout<<"Matched Words: ";
            int rk=rabinKarp(line);

            clock_t end1=clock();

            double time1=double(end1-start1)/CLOCKS_PER_SEC;

            cout<<endl;
            cout<<"Total Matches: "<<rk<<endl;
            cout<<"Time Taken: "<<time1<<" sec"<<endl;

            //aho corasick
            clock_t start2=clock();

            int ac=ahoSearch(line);

            clock_t end2=clock();

            double time2=double(end2-start2)/CLOCKS_PER_SEC;

            cout<<"Aho Corasick Total Matches: "<<ac<<endl;
            cout<<"Time Taken: "<<time2<<" sec"<<endl;

            if(rk>0 || ac>0)
```

```
                cout<<"SPAM MESSAGE!! CAUTION !!"<<endl;
            else
                cout<<"NORMAL MESSAGE!"<<endl;

            msgNo++;

            if(msgNo>20)
                break;
        }

        file.close();

        return 0;
    }
```

⇨ ScreenShot:

Welcome | daaproj.cpp

C: > sem4 projects > daa > daaproj.cpp > main()

```cpp
#include <bits/stdc++.h>
using namespace std;

//50 spam keywords taken from the spam.csv dataset

string patterns[50]={
"win money","free offer","click here","claim prize","urgent",
"limited time","loan approved","cash reward","work from home","be your own boss",
"exclusive deal","call now","reply now","free gift","earn money",
"winner","congratulations","special offer","risk free","bonus",
"double money","act now","hurry up","subscribe now","buy now",
"discount","selected","guaranteed","cheap rate","investment",
"lottery","reward now","free cash","easy income","no cost",
"claim now","instant loan","extra income","offer ends","text now",
"visit now","password reset","verify account","bank details","miracle cure",
"no obligation","gift card","trial offer","free entry","jackpot"
};

string toLower(string s){
    for(int i=0; i<s.length(); i++)
        s[i]=tolower(s[i]);
    return s;
}

//search using RabinKarp
int rabinKarp(string text){
    int i;
    int count=0;

    text=toLower(text);

    for(i=0; i<50; i++){
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | PORTS

DEBUG CONSOLE | TERMINAL | CHAT

```
PS C:\Users\Vishwajeet> cd 'c:\sem4 projects\daa\output'
PS C:\sem4 projects\daa\output> & .\'daaproj.exe'
C VISHWAJEET KUMAR
24BCE0341
CS-1 REAL TIME SPAM DETECTION SYSTEM
```

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

Compile & Run | Ln 222, Col 2 | Spaces: 4 | UTF-8 | CRLF | C++ | Go Live | Win32

Welcome | daaproj.cpp

C: > sem4 projects > daa > daaproj.cpp > main()

```cpp
string toLower(string s){

//search using RabinKarp
int rabinKarp(string text){
    int i;
    int count=0;

    text=toLower(text);

    for(i=0; i<50; i++){
        if(text.find(patterns[i])!=string::npos){
            cout<<patterns[i]<<", ";
            count++;
        }
    }

    return count;
}

//aho corasick
struct Node{
    int next[26];
    int fail;
    int output;

    Node(){
        int i;

        for(i=0; i<26; i++)
            next[i]=-1;

        fail=0;
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | PORTS

DEBUG CONSOLE | TERMINAL | CHAT

```
PS C:\Users\Vishwajeet> cd 'c:\sem4 projects\daa\output'
PS C:\sem4 projects\daa\output> & .\'daaproj.exe'
C VISHWAJEET KUMAR
24BCE0341
CS-1 REAL TIME SPAM DETECTION SYSTEM
```

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

Compile & Run | Ln 222, Col 2 | Spaces: 4 | UTF-8 | CRLF | C++ | Go Live | Win32

Welcome | daaproj.cpp

C: > sem4 projects > daa > daaproj.cpp > main()

```cpp
struct Node{
    Node(){
};

Node trie[3000];
int sizeTrie=1;

//putting in pattern
void insertWord(string s){
    int node=0;
    int i, x;

    for(i=0; i<s.length(); i++){

        if(s[i]==' ')
            continue;

        if(s[i]<'a' || s[i]>'z')
            continue;

        x=s[i]-'a';

        if(trie[node].next[x]==-1){
            trie[node].next[x]=sizeTrie;
            sizeTrie++;
        }

        node=trie[node].next[x];
    }

    trie[node].output++;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

DEBUG CONSOLE | TERMINAL | CHAT

```
PS C:\Users\Vishwajeet> cd 'c:\sem4 projects\daa\output'
PS C:\sem4 projects\daa\output> & .\'daaproj.exe'
C VISHWAJEET KUMAR
24BCE0341
CS-1 REAL TIME SPAM DETECTION SYSTEM
```

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

Compile & Run | Ln 222, Col 2 | Spaces: 4 | UTF-8 | CRLF | C++ | Go Live | Win32

Welcome | daaproj.cpp

C: > sem4 projects > daa > daaproj.cpp > main()

```cpp
void insertWord(string s){
}

//building the trie
void buildTrie(){
    queue<int> q;
    int i, u, v;

    for(i=0; i<26; i++){

        if(trie[0].next[i]!=-1){
            trie[trie[0].next[i]].fail=0;
            q.push(trie[0].next[i]);
        }
        else{
            trie[0].next[i]=0;
        }
    }

    while(!q.empty()){

        u=q.front();
        q.pop();

        for(i=0; i<26; i++){

            v=trie[u].next[i];

            if(v!=-1){
                trie[v].fail=trie[trie[u].fail].next[i];
                q.push(v);
            }
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

DEBUG CONSOLE | TERMINAL | CHAT

```
PS C:\Users\Vishwajeet> cd 'c:\sem4 projects\daa\output'
PS C:\sem4 projects\daa\output> & .\'daaproj.exe'
C VISHWAJEET KUMAR
24BCE0341
CS-1 REAL TIME SPAM DETECTION SYSTEM
```

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

Compile & Run | Ln 222, Col 2 | Spaces: 4 | UTF-8 | CRLF | C++ | Go Live | Win32

Welcome | daaproj.cpp

C: > sem4 projects > daa > daaproj.cpp > main()

```cpp
void buildTrie(){
    while(!q.empty()){
        for(i=0; i<26; i++){

    }
}

//search text
int ahoSearch(string text){

    int node=0;
    int i, x;
    int count=0;

    text=toLower(text);

    for(i=0; i<text.length(); i++){

        if(text[i]==' ')
            continue;

        if(text[i]<'a' || text[i]>'z')
            continue;

        x=text[i]-'a';

        node=trie[node].next[x];

        count=count+trie[node].output;
    }

    return count;
}
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | PORTS

DEBUG CONSOLE | TERMINAL | CHAT

```
PS C:\Users\Vishwajeet> cd 'c:\sem4 projects\daa\output'
PS C:\sem4 projects\daa\output> & .\'daaproj.exe'
C VISHWAJEET KUMAR
24BCE0341
CS-1 REAL TIME SPAM DETECTION SYSTEM
```

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

Compile & Run | Ln 222, Col 2 | Spaces: 4 | UTF-8 | CRLF | C++ | Go Live | Win32

Welcome | daaproj.cpp

C: > sem4 projects > daa > daaproj.cpp > main()

```cpp
int ahoSearch(string text){
}

//main
int main(){

    cout<<"C VISHWAJEET KUMAR"<<endl;
    cout<<"24BCE0341"<<endl;
    cout<<"CS-1 REAL TIME SPAM DETECTION SYSTEM"<<endl;
    cout<<"RABIN KARP VS AHO-CORASICK"<<endl;

    int i=0;
    string line;

    ifstream file("C:\\sem4 projects\\spam.csv");

    if(!file){
        cout<<"File not found"<<endl;
        return 0;
    }

    //building trie
    for(i=0; i<50; i++)
        insertWord(toLower(patterns[i]));

    buildTrie();

    int msgNo=1;

    while(getline(file,line))
    {
        cout<<"\nMessage "<<msgNo<<endl;
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | PORTS

DEBUG CONSOLE | TERMINAL | CHAT

```
PS C:\Users\Vishwajeet> cd 'c:\sem4 projects\daa\output'
PS C:\sem4 projects\daa\output> & .\'daaproj.exe'
C VISHWAJEET KUMAR
24BCE0341
CS-1 REAL TIME SPAM DETECTION SYSTEM
```

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

Compile & Run | Ln 222, Col 2 | Spaces: 4 | UTF-8 | CRLF | C++ | Go Live | Win32

```cpp
    while(getline(file,line))
    {
        cout<<"\nMessage "<<msgNo<<endl;
        cout<<"Text: "<<line<<endl;

        //measuring time for rabin karp
        clock_t start1=clock();

        cout<<"Matched Words: ";
        int rk=rabinKarp(line);

        clock_t end1=clock();

        double time1=double(end1-start1)/CLOCKS_PER_SEC;

        cout<<endl;
        cout<<"Total Matches: "<<rk<<endl;
        cout<<"Time Taken: "<<time1<<" sec"<<endl;

        //aho corasick
        clock_t start2=clock();

        int ac=ahoSearch(line);

        clock_t end2=clock();

        double time2=double(end2-start2)/CLOCKS_PER_SEC;

        cout<<"Aho Corasick Total Matches: "<<ac<<endl;
        cout<<"Time Taken: "<<time2<<" sec"<<endl;

        if(rk>0 || ac>0)
```

```
Message 1
Text: v1,v2,,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
```

```cpp
        int ac=ahoSearch(line);

        clock_t end2=clock();

        double time2=double(end2-start2)/CLOCKS_PER_SEC;

        cout<<"Aho Corasick Total Matches: "<<ac<<endl;
        cout<<"Time Taken: "<<time2<<" sec"<<endl;

        if(rk>0 || ac>0)
            cout<<"SPAM MESSAGE!! CAUTION !!"<<endl;
        else
            cout<<"NORMAL MESSAGE!"<<endl;

        msgNo++;

        if(msgNo>20)
            break;
    }

    file.close();

    return 0;
}
```

```
Message 1
Text: v1,v2,,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
```

OUTPUT:

Welcome | daaproj.cpp

C: > sem4 projects > daa > daaproj.cpp > main()

```cpp
int ac=ahoSearch(line);

clock_t end2=clock();
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | PORTS

DEBUG CONSOLE

Filter (e.g. text, !exclude, \...

TERMINAL

```
PS C:\Users\Vishwajeet> cd 'c:\sem4 projects\daa\output'
C VISHWAJEET KUMAR
24BCE0341
CS-1 REAL TIME SPAM DETECTION SYSTEM
RABIN KARP VS AHO-CORASICK

Message 1
Text: v1,v2,,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 2
Text: ham,"Go until jurong point, crazy.. Available only in bugis n grea
t world la e buffet... Cine there got amore wat...",,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 3
Text: ham,Ok lar... Joking wif u oni...,,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!
```

CHAT

SESSIONS

Error trouble
4 hrs ago

Retrieving las
2 mos ago

Tip: Try the Plan
changes.

daaproj.cp

Describe what

Agent

Local

Welcome | daaproj.cpp

C: > sem4 projects > daa > daaproj.cpp > main()

```cpp
int ac=ahoSearch(line);

clock_t end2=clock();
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | PORTS

DEBUG CONSOLE | TERMINAL

```
Message 4
Text: spam,Free entry in 2 a wkly comp to win FA Cup final tkts 21st May
 2005. Text FA to 87121 to receive entry question(std txt rate)T&C's app
ly 08452810075over18's,,,
Matched Words: free entry,
Total Matches: 1
Time Taken: 0.002 sec
Aho Corasick Total Matches: 1
Time Taken: 0 sec
SPAM MESSAGE!! CAUTION !!

Message 5
Text: ham,U dun say so early hor... U c already then say...,,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 6
Text: ham,"Nah I don't think he goes to usf, he lives around here though
",,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 7
Text: spam,"FreeMsg Hey there darling it's been 3 week's now and no word
 back! I'd like some fun you up for it still? Tb ok! XxX std chgs to sen
```

Compile & Run | Ln 222, Col 2

Welcome | daaproj.cpp

C: > sem4 projects > daa > daaproj.cpp > main()

```
198
199        int ac=ahoSearch(line);
200
201        clock_t end2=clock();
202
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | PORTS

DEBUG CONSOLE

Filter (e.g. text, !exclude, \...

TERMINAL

```
Message 7
Text: spam,"FreeMsg Hey there darling it's been 3 week's now and no word
 back! I'd like some fun you up for it still? Tb ok! XxX std chgs to sen
d, σú1.50 to rcv",,,
Matched Words:
Total Matches: 0
Time Taken: 0.001 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 8
Text: ham,Even my brother is not like to speak with me. They treat me li
ke aids patent.,,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 9
Text: ham,As per your request 'Melle Melle (Oru Minnaminunginte Nurungu
Vettam)' has been set as your callertune for all Callers. Press *9 to co
py your friends Callertune,,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 10
Text: spam,WINNER!! As a valued network customer you have been selected
```

CHAT

SESSIONS

Error trou
4 hrs ago

Retrieving
2 mos ago

Tip: Try the
changes.

daapr

Describe w

Local

Compile & Run

Ln 222, Col 2 | Spaces: 4

Welcome | daaproj.cpp

C: > sem4 projects > daa > daaproj.cpp > main()

```
198
199         int ac=ahoSearch(line);
200
201         clock_t end2=clock();
202
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | PORTS

DEBUG CONSOLE

Filter (e.g. text, !exclude, \...

TERMINAL

```
Message 11
Text: spam,Had your mobile 11 months or more? U R entitled to Update to
the latest colour mobiles with camera for Free! Call The Mobile Update C
o FREE on 08002986030,,,
Matched Words:
Total Matches: 0
Time Taken: 0.001 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 12
Text: ham,"I'm gonna be home soon and i don't want to talk about this st
uff anymore tonight, k? I've cried enough today.",,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 13
Text: spam,"SIX chances to win CASH! From 100 to 20,000 pounds txt> CSH1
1 and send to 87575. Cost 150p/day, 6days, 16+ TsandCs apply Reply HL 4
info",,,
Matched Words:
Total Matches: 0
Time Taken: 0.001 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 14
```

CHAT

SESSIONS

Error tr
4 hrs ag

Retrievi
2 mos a

Tip: Try th
changes.

Describe

Local

Compile & Run

Ln 222, Col 2 Spaces:

Welcome | daaproj.cpp

C: > sem4 projects > daa > daaproj.cpp > main()

```
198
199         int ac=ahoSearch(line);
200
201         clock_t end2=clock();
202
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | PORTS

DEBUG CONSOLE

Filter (e.g. text, !exclude, \...

TERMINAL

```
Message 14
Text: spam,"URGENT! You have won a 1 week FREE membership in our oú100,0
00 Prize Jackpot! Txt the word: CLAIM to No: 81010 T&C www.dbuk.net LCCL
TD POBOX 4403LDNW1A7RW18",,,
Matched Words: urgent, jackpot,
Total Matches: 2
Time Taken: 0.001 sec
Aho Corasick Total Matches: 2
Time Taken: 0 sec
SPAM MESSAGE!! CAUTION !!

Message 15
Text: ham,I've been searching for the right words to thank you for this
breather. I promise i wont take your help for granted and will fulfil my
 promise. You have been wonderful and a blessing at all times.,,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 16
Text: ham,I HAVE A DATE ON SUNDAY WITH WILL!!,,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 17
Text: spam,"XXXMobileMovieClub: To use your credit, click the WAP link i
n the next txt message or click here>> http://wap. xxxmobilemovieclub.co
```

CHAT

SESSIONS

Error trou
4 hrs ago

Retrievin
2 mos ag

Tip: Try the
changes.

daap

Describe

Local

Compile & Run

Ln 222, Col 2 | Spaces: 4

Welcome | daaproj.cpp

C: > sem4 projects > daa > daaproj.cpp > main()

```
int ac=ahoSearch(line);

clock_t end2=clock();
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | PORTS

DEBUG CONSOLE | TERMINAL

```
Message 17
Text: spam,"XXXMobileMovieClub: To use your credit, click the WAP link i
n the next txt message or click here>> http://wap. xxxmobilemovieclub.co
m?n=QJKGIGHJJGCBL",,,
Matched Words: click here,
Total Matches: 1
Time Taken: 0 sec
Aho Corasick Total Matches: 1
Time Taken: 0 sec
SPAM MESSAGE!! CAUTION !!

Message 18
Text: ham,Oh k...i'm watching here:),,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 19
Text: ham,Eh u remember how 2 spell his name... Yes i did. He v naughty
make until i v wet.,,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!

Message 20
Text: ham,Fine if thatσ┌s the way u feel. Thatσ┌s the way its gota b,,,
Matched Words:
Total Matches: 0
```

Compile & Run | Ln 222, Col 2

```
NORMAL MESSAGE!

Message 20
Text: ham,Fine if thatσ┌s the way u feel. Thatσ┌s the way its gota b,,,
Matched Words:
Total Matches: 0
Time Taken: 0 sec
Aho Corasick Total Matches: 0
Time Taken: 0 sec
NORMAL MESSAGE!
PS C:\sem4 projects\daa\output> cd 'c:\sem4 projects\daa\output'
PS C:\sem4 projects\daa\output> & .\'daaproj.exe'
C VISHWAJEET KUMAR
```

7. Analysis:

7.1 Performance vs Number of Patterns:

When the number of spam keywords increases, Rabin Karp becomes slower because it checks each pattern separately. Aho Corasick remains faster because it searches all patterns together in one scan. Therefore, Aho-Corasick is better for large number of Pattern.

7.2 Performance vs Text Size:

For small messages, both algorithms work fast. For large messages such as long emails, Rabin Karp takes more time, while Aho-Corasick remains efficient because it scans the text only once. Hence, Aho-Corasick performs better for large text.

7.3 Real Time Behaviour:

In real time spam filtering, messages must be checked instantly. Rabin Karp may slow down when many keywords are used. Aho Corasick gives faster response time and is suitable for live systems.

7.4 Memory Usage:

Rabin karp is simple and memory efficient but aho corasick is faster and better for real time spam detection

8. Conclusion and Inference:

Conclusion:

- This project successfully implemented a Real Time Spam Detection System using Rabin–Karp and Aho–Corasick algorithms.
- Both algorithms were able to identify predefined spam keywords from incoming messages such as emails, SMS, and chat texts.
- Rabin–Karp performed well when the number of keywords and message size were small. It is simple to implement and uses less memory.
- As the number of spam patterns increased from 50 to 500, Rabin–Karp required more processing time because each pattern was searched separately.
- Aho–Corasick showed better efficiency by matching multiple patterns simultaneously in a single scan of the text.
- It provided faster execution time for large messages and larger keyword sets.
- Although Aho–Corasick uses more memory due to trie and failure links, its speed advantage makes it highly suitable for practical systems.
- Therefore, for modern spam filtering applications where fast response is important, Aho–Corasick is the better choice.

Inference:

- For small-scale systems with limited keywords, Rabin–Karp can be used effectively.
- For large-scale spam detection with hundreds of keywords, Aho–Corasick performs much better.
- If memory resources are limited, Rabin–Karp is preferable.
- If speed and real-time performance are the main priorities, Aho–Corasick is the ideal algorithm.
- Increasing text size affects Rabin–Karp more than Aho–Corasick.

- Increasing number of patterns has minimal effect on Aho–Corasick after trie construction.
- Aho–Corasick is more scalable for future expansion of spam keyword databases.
- Real-world email and messaging platforms would benefit more from Aho–Corasick based filtering systems.
- The project proves that algorithm selection depends on system requirements such as speed, memory, and data size.
- Overall, Aho–Corasick is recommended for real-time spam detection systems.

PS: GITHUB LINK:
https://github.com/wejew2983/DAA_project_SpamDetection